

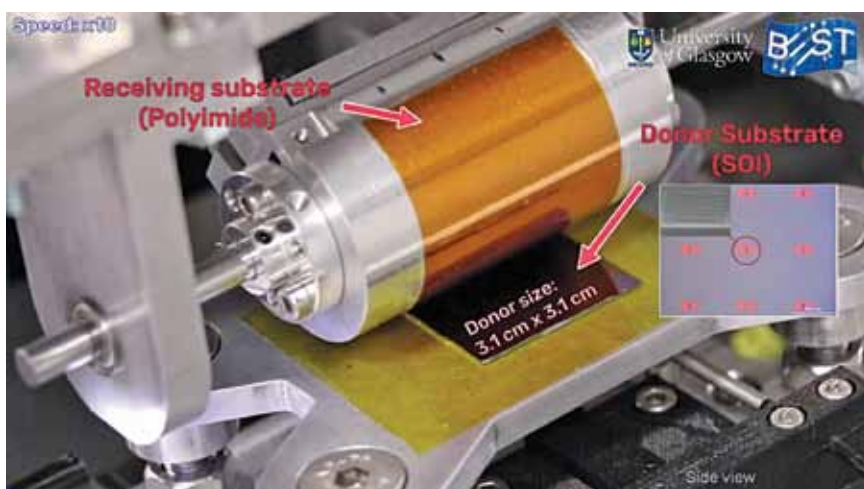


density even at the smallest sizes. They developed a new kind of current collector and cathode that increases the energy storage capacity while serving as a protective shell. This arrangement reduces the need for non-conductive packaging that normally protects a battery's sensitive internal chemicals.

Weighing about as much as two grains of rice but with the energy density of a much larger, heavier battery, the researchers' packing-free design could enable a host of otherwise impossible electronics (Credit: <https://penntoday.upenn.edu>)

New method for fabricating flexible electronics

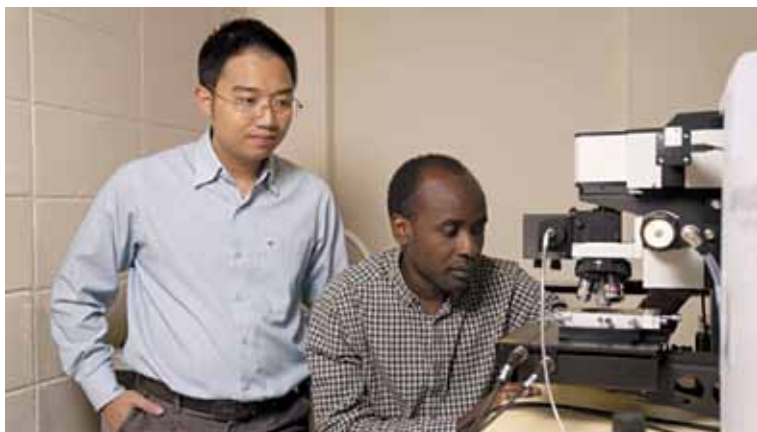
Until now, flexible electronics have been mainly manufactured by a process called transfer printing, a three-stage stamping process a bit like receiving an ink stamp. However, the process has a number of limitations that have made it challenging to create more large-scale, complex flexible devices. Engineers from the University of Glasgow's Bendable Electronics and Sensing Technologies have now improved the conventional approach for creating flexible electronics. Instead of transferring nanostructures to a soft polymeric stamp before it is transferred to the final substrate, their new process called 'direct roll transfer' prints silicon straight onto a flexible surface. The prepared substrate is wrapped around a metal tube, and a computer-controlled machine



Direct roll printing for high performance electronics (Credit: www.gla.ac.uk)

then rolls the tube over the silicon wafer, transferring it to the flexible material.

Ultra-thin 2D materials found to produce electricity by twisting



A new study from The Australian National University shows the ability of 2D materials to convert sunlight into electricity by simply 'twisting' the angle between two ultra-thin layers correctly. The new class of materials (2D) are 100,000 times thinner than a single sheet of paper and could be used in a huge range of technology, including solar cells, LED lights, and sensing devices. Two different 2D materials are stacked together to move positive and negative charges in opposite directions, generating electricity.

Twist brings new possibilities for ultra-thin 2D materials (Credit: www.anu.edu.au)